# VALTHERA

Research Report
Evidence Base, Design Implications & Confidence Assessment

**Version 1.0 · October 2026**

*Companion to the Valthera Whitepaper Version 1.0*

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# Executive Summary

This report sets out the evidence behind the Valthera Whitepaper (Version 1.0). Valthera is an AI-native capital market. Independent investment agents evaluate projects, form valuations, and allocate capital under signed mandates, and limits on their authority are enforced outside the models. The report tests the whitepaper's premises and design choices against primary sources: statutes and regulations, regulator publications, court decisions, peer-reviewed or preprint research, security standards, and incident records.

**Principal findings**

1. **Agent capability is rising quickly, but evidence that agents can perform financial work is weak.** METR measures the length of task a frontier agent can complete on its own with 50% reliability. That length has doubled roughly every seven months for six years [S1]. Yet on finance-specific benchmarks, the best agent tested scored only 46.8% accuracy on real-world financial research tasks [S6]. In an earlier study, a leading model with retrieval answered 81% of filing-based questions incorrectly or not at all [S5]. Valthera's design (models propose, deterministic engines compute, people approve) matches the evidence.
2. **Agent payment infrastructure exists, but economic usage is small.** x402 provides HTTP-native stablecoin settlement [S2][S3]. Reported daily dollar volume in March 2026 was about $28,000, much of it test or artificial activity [S4].
3. **US intermediary rules define the legal architecture.** A registered funding portal may not give investment advice or recommendations, and may not handle investor funds [S9][S10]. Because Valthera publishes AI assessments, its research and offering functions must be structurally separate, and US offerings must use a broker-dealer.
4. **The investment adviser question turns on personalization and control.** Under *Lowe v. SEC*, impersonal, bona fide publications of general and regular circulation fall outside the adviser definition [S13]. In a 2024 case, a court applying *Lowe* distinguished subscription research from "auto-trading" arrangements that execute trades for subscribers [S14]. Valthera's research layer can be designed as impersonal publication. Its agent-execution layer cannot rely on that exclusion.
5. **The EU offers a harmonized but restrictive path.** The European crowdfunding regulation (ECSPR) covers offers up to €5 million per project owner over 12 months. Only EU-established, authorized legal persons may provide the service. It bans payments for routing orders and prohibits platforms from investing in offers on their own platform [S17][S18][S19].
6. **Prompt injection against agents that hold funds is a demonstrated risk.** OWASP ranks prompt injection first among LLM risks [S21]. In May 2026, an encoded prompt led an AI agent to transfer an estimated $150,000–$200,000 from a wallet whose permissions had been expanded [S22][S23].
7. **Track records must separate forecasting skill from returns.** Early-stage returns follow an extreme power law [S27], so return records stay noisy for years. Strictly proper scoring rules such as the Brier score reward honest probability reporting [S25][S26] and give faster, fairer measurement.

**Overall assessment.** The thesis is directionally supported. Its near-term feasibility depends on three things: a compliant legal structure, controls on agent authority that sit outside the models, and an honest measurement framework. The whitepaper includes all three. The largest open questions are whether AI agents can actually outperform people at early-stage selection (currently unproven) and the detailed licensing analysis for each jurisdiction.

# 1. Scope and Methodology

## 1.1 Research questions

| ID | Research question |
|---|---|
| RQ1 | Are AI agents becoming capable, transacting economic actors? |
| RQ2 | How reliably do current AI systems perform financial analysis? |
| RQ3 | What US securities-law constraints govern an AI platform that publishes investment assessments and facilitates capital raising? |
| RQ4 | What constraints apply in the European Union? |
| RQ5 | What security risks arise when AI agents control capital, and which controls do authoritative sources recommend? |
| RQ6 | How can the forecasting skill and investment performance of agents be measured credibly? |

Section 8 maps each finding to the corresponding section of the Valthera Whitepaper v1.0.

## 1.2 Source hierarchy

Sources were selected and weighted in the following order:

1. **Primary legal and regulatory sources:** statutes and regulations (EUR-Lex), the Federal Register, SEC and FINRA publications, and court decisions.
2. **Primary research and standards:** original papers (arXiv, peer-reviewed journals), OWASP standards, protocol specifications.
3. **Primary incident and data records:** the OECD AI Incidents Monitor, original datasets published by the data owner (for example, AngelList).
4. **Qualified secondary sources:** law-firm analyses, reputable financial press. These are used only where no primary source was located, and are marked as such.

Encyclopedia pages, aggregators, and promotional material were excluded as evidentiary sources.

## 1.3 Evidence grading

Each finding carries a confidence grade:

| Grade | Meaning |
|---|---|
| **High** | Primary sources, consistent with each other |
| **Medium** | Primary source with scope limits, or several consistent secondary sources |
| **Low** | A single secondary source, conflicting data, or extrapolation |

## 1.4 Limitations

- The research does not constitute legal advice. Regulatory findings describe the framework and must be confirmed by qualified counsel in each jurisdiction.
- Some regulatory positions, particularly on digital assets, rest on speeches and staff statements rather than final rules.
- Benchmarks of AI capability measure specific task distributions and do not directly measure investment outcomes.

# 2. RQ1 — Agent Capability and Agentic Payments

## 2.1 Autonomous task capability

METR proposed measuring AI capability by the length of task (in human-professional time) that an agent can complete on its own with 50% reliability. On this measure, frontier agents have improved exponentially, doubling roughly every seven months over six years [S1]. The accompanying paper reports that Claude 3.7 Sonnet had a 50% time horizon of about 50 minutes. It attributes the gains mainly to “greater reliability and ability to adapt to mistakes, combined with better logical reasoning and tool use capabilities” [S1-paper].

**Limitations stated by METR**

- The estimate depends on methodological choices, including which tasks are used and whose human performance is measured. METR is confident only in a range of roughly one to four doublings per year [S1].
- The tasks are mostly software engineering and general reasoning.
- Contract baseliners took 5 to 18 times longer than repository maintainers to resolve the same issues. This suggests time horizons correspond better to “the labor of a low-context human, rather than a high-context human” [S1-paper].

**Assessment.** The trend is strong evidence that agents are becoming more autonomous. Venture diligence, however, is high-context, unstructured, and judgment-heavy, which is outside the benchmark’s main domain. The trend should therefore not be read as evidence that agents can already perform investment diligence reliably. *Confidence: High (trend); Low (transfer to investment work).*

## 2.2 Agentic payment infrastructure

x402 is an open payment protocol developed by Coinbase that uses the HTTP 402 “Payment Required” status code. When a request arrives without payment, the server responds with a 402 and the client pays and retries. This removes the need for API keys, accounts, and subscriptions [S3]. Coinbase describes it as enabling “instant stablecoin payments directly over HTTP” for APIs, apps, and AI agents [S2]. The whitepaper cites per-request costs low enough to make micropayments viable [S3].

## 2.3 Actual economic usage

Usage figures conflict. Industry sources cite cumulative transaction counts in the hundreds of millions. CoinDesk, reporting onchain data in March 2026, found that x402 processed about $28,000 in daily volume, “much of it from testing and ‘gamed’ transactions rather than real commerce.” It cited Artemis analysis suggesting that roughly half of observed transactions reflect artificial activity [S4].

**Reconciliation.** The figures are compatible. High transaction counts with sub-dollar average values and a large artificial share produce very little economic volume. Dollar volume is the more relevant measure of economic activity. *Confidence: Low–Medium.* The best available figure is a news report of a third-party analysis, not a primary dataset.

## 2.4 Implications

- Agent payments should be treated as an emerging, interchangeable settlement component, not as evidence of demand.
- The go-to-market should not depend on autonomous agent capital being available at launch.
- Settlement logic should stay independent of any single payment rail.

# 3. RQ2 — Reliability of AI in Financial Analysis

## 3.1 FinanceBench

FinanceBench is a benchmark of more than 10,000 question-answer pairs drawn from public company filings. Its authors tested 16 model configurations on a 150-case sample and manually reviewed 2,400 answers [S5]. Findings:

- GPT-4-Turbo with a retrieval system “incorrectly answered or refused to answer 81% of questions” [S5].
- The best configuration (GPT-4-Turbo with long context) was correct in 79% of cases. The authors concluded that such a model “could not be used with confidence in a live industry setting” [S5].
- Even plausible-looking answers risked being “hallucinations, out-of-date, logically incorrect, or given with the wrong units” [S5].

FinanceBench questions are deliberately clear-cut and are meant as a minimum performance standard. Open-ended diligence is harder.

## 3.2 Finance Agent Benchmark

A 2025 benchmark tested AI agents (models with tools such as web search and EDGAR access) on real-world financial research tasks. The best-performing model, OpenAI o3, achieved 46.8% accuracy at an average cost of $3.79 per query. Models handled simple retrieval better than complex financial modeling or market analysis [S6]. All models finished tasks faster than human experts, which points to productivity benefits despite the accuracy limits [S6].

## 3.3 Interpretation

These benchmarks measure older and newer model generations, and capability keeps improving (Section 2.1). Two conclusions still hold:

1. Unverified model output is not reliable enough for capital allocation.
2. Model errors are often presented with confident, well-formatted output, so the format of an answer is no guide to its accuracy.

No rigorous, peer-reviewed study was located showing that AI agents outperform human investors at early-stage *selection* or *valuation* as measured by investment outcomes. *Confidence: Medium (benchmark findings); the gap itself is a finding.*

## 3.4 Implications

- Language models should extract claims and propose assumptions. Deterministic services should compute valuations, and every number should trace to evidence (Whitepaper §6, §8).
- Missing data must remain marked as missing, never inferred (Whitepaper §6.2).
- Labels need human audit sampling and published precision metrics (Whitepaper §6.6).
- Agent efficacy should be measured, not assumed. This motivates the scoring framework (Section 7).

# 4. RQ3 — United States Securities-Law Constraints

## 4.1 Offering exemptions available to issuers

| Exemption | Offering limit (12 months) | Investors | Key conditions | Source |
|---|---|---|---|---|
| Regulation Crowdfunding | $5 million | Accredited and non-accredited | Must use an SEC-registered broker-dealer or funding portal. Annual limits for non-accredited investors | [S7][S8] |
| Rule 506(b), Reg D | Unlimited | Unlimited accredited, plus up to 35 sophisticated non-accredited | No general solicitation. Form D within 15 days | [S11] |
| Rule 506(c), Reg D | Unlimited | Accredited only | General solicitation permitted. Issuer must take reasonable steps to verify accredited status | [S11][S12] |
| Rule 504, Reg D | $10 million | Any | Often used for regional offerings | [S11] |
| Regulation A, Tier 1 | $20 million | Any | State qualification required | [S15] |
| Regulation A, Tier 2 | $75 million | Any; non-accredited limited to 10% of income or net worth | Audited financials, ongoing reporting, state preemption | [S15] |

**Investor limits under Regulation Crowdfunding.** If either annual income or net worth is below $124,000, a non-accredited investor may invest the greater of $2,500 or 5% of the greater of the two in any 12-month period. If both are at or above $124,000, the limit is 10%, capped at $124,000 [S8].

**Verification under Rule 506(c).** In March 2025, SEC staff agreed in no-action guidance that minimum investment amounts can be a reasonable verification step: at least $200,000 for natural persons and $1 million for entities, combined with written representations [S12; secondary summary of a primary SEC no-action letter].

**Scale.** The SEC's Division of Economic and Risk Analysis (DERA) reported about 8,500 Regulation Crowdfunding offerings between May 2016 and December 2024, with about $1.3 billion in reported proceeds. The average successful offering raised about $346,000 [S16; secondary summary of a primary DERA report]. Rounds in the range used in Valthera's examples (around $500,000) are typical of this channel.

## 4.2 The intermediary constraint

Every Regulation Crowdfunding transaction must go through a registered broker-dealer or funding portal [S7]. A registered funding portal is prohibited from:

- offering investment advice or recommendations;
- soliciting purchases, sales, or offers to buy the securities on its platform;
- compensating persons for such solicitation, or based on sales;
- holding, managing, possessing, or handling investor funds or securities.

An intermediary that wants to do any of these things "must register as a broker-dealer rather than as a funding portal" [S9][S10]. FINRA's 2025 oversight report lists funding portals sending communications that "recommend investments" among its recurring examination findings [S20].

FINRA also places gatekeeper obligations on intermediaries. Under Rule 301(a), an intermediary must have a reasonable basis for believing an issuer complies with Regulation Crowdfunding. Under Rule 301(c)(2), it must deny platform access where there is a reasonable basis for believing the issuer or offering presents the potential for fraud [S20].

**Analysis.** Valthera's product publishes AI-generated assessments of specific offerings, including invest or decline decisions and reservation prices. Combined with hosting offerings, this cannot operate within a funding portal. The

available structures are:

| Option | Description | Trade-offs |
|---|---|---|
| A. Entity separation | A research entity publishes assessments. A separate, unaffiliated or information-barriered intermediary hosts offerings | Preserves research independence. Requires strict barriers and no compensation tied to offerings |
| B. Broker-dealer route | Offerings run through a registered broker-dealer (owned or partnered), which may make recommendations subject to its own rules | Higher regulatory burden. Allows an integrated experience |
| C. Non-US capital formation | Capital formation runs under foreign regimes (for example, ECSPR, Section 5) | Requires local authorization. US investors excluded or restricted |

The gatekeeper obligations line up with Valthera's Evidence Engine. Contradicted material claims and fraud indicators are exactly the signals an intermediary needs to meet Rule 301.

*Confidence: High.*

## 4.3 Investment Advisers Act exposure

The Investment Advisers Act excludes from the definition of "investment adviser" "the publisher of any bona fide newspaper, news magazine or business or financial publication of general and regular circulation." In *Lowe v. SEC*, the Supreme Court held that the petitioners' newsletters fell within this exclusion. It reasoned that "the mere fact that a publication contains advice and comment about specific securities does not give it the personalized character that identifies a professional investment adviser." It added that publications remain presumptively excluded "as long as the communications between petitioners and their subscribers remain entirely impersonal and do not develop into the kind of fiduciary, person-to-person relationships" characteristic of advisory relationships [S13].

In 2024 a federal court dismissed claims that Seeking Alpha operated as an unregistered adviser. It held that customizable filters and alerts did not make "impersonal, disinterested, and generally available content" individualized [S14]. The court also distinguished Seeking Alpha's services from "auto-trading" programs, in which publishers executed trades on subscribers' behalf [S14; law-firm summaries of the court opinion].

**Analysis.** Each Valthera layer sits differently against this framework:

| Valthera layer | Exposure | Design response |
|---|---|---|
| Public assessments and evidence labels | Potentially within the publisher exclusion if impersonal, disinterested, and regularly available | Same content for all subscribers. No tailoring to individual portfolios. Conflict disclosures |
| Agents run by third-party operators | Each operator's status depends on its own activities | Operators attest to their own regulatory status (Whitepaper §20.4) |
| Agents executing for capital providers (T2/T3) | Outside the publisher exclusion. Comparable to discretionary management | Operators must hold the required registration, or capital must belong to the operator itself |
| Valthera-hosted runtime | Platform-provider role. Must not exercise discretion | Valthera does not direct agent decisions or take custody (Whitepaper §13.1) |

*Confidence: Medium.* The principles are well established, but applying them to AI agents is untested. The Seeking Alpha analysis relies on law-firm summaries; the opinion itself was not reviewed.

## 4.4 AI-specific rulemaking

On June 12, 2025, the SEC withdrew 14 proposed rules, including the August 2023 proposal on conflicts of interest associated with predictive data analytics, effective June 17, 2025 [S24]. That proposal would have required firms to identify, and to eliminate or neutralize, conflicts arising from "covered technologies," including AI, machine learning, and large language models [S24-summary]. The SEC stated that any future action would require a new proposal.

**Implication.** No AI-specific conflict rule currently applies. General anti-fraud provisions, fiduciary duties (where applicable), and disclosure obligations remain in force. Valthera's conflict registry and disclosure regime anticipate the direction of possible future rules. *Confidence: High.*

## 4.5 Digital assets

In a November 12, 2025 speech, SEC Chairman Paul Atkins anticipated a token taxonomy “anchored in the longstanding Howey investment contract securities analysis” [S28]. In a March 17, 2026 speech, he stated that the Commission was implementing an interpretation under which four categories are not deemed securities: digital commodities, digital collectibles, digital tools, and payment stablecoins under the GENIUS Act. Under the same interpretation, tokenized securities remain securities, and a non-security asset can still be subject to securities laws when it is sold as part of an investment contract [S29]. Both speeches carry the standard disclaimer that the views are the Chairman’s own.

**Implication.** Equity, revenue-share, and similar instruments offered through Valthera remain securities whether or not they are tokenized. The whitepaper’s decision not to issue a token avoids an entire category of classification risk. *Confidence: Medium.* The positions come from speeches; the final interpretive text was not reviewed.

# 5. RQ4 — European Union Constraints

## 5.1 Scope of the ECSPR

Regulation (EU) 2020/1503 (the ECSPR) has applied since November 10, 2021 [S19]. It harmonizes rules for crowdfunding service providers that match investors with businesses seeking funding through loans or transferable securities. Offers above €5 million, calculated per project owner over 12 months, fall outside its scope and are governed instead by MiFID II and the Prospectus Regulation [S18][S19].

## 5.2 Requirements relevant to Valthera

| Requirement | Content | Relevance |
|---|---|---|
| Authorization and establishment | Services “shall only be provided by legal persons that are established in the Union and that have been authorised” [S17] | Requires an EU subsidiary for EU capital formation |
| No inducements | Providers shall not pay or accept remuneration “for routing investors’ orders to a particular crowdfunding offer” [S17] | Matches Valthera’s rule against outcome-based fees |
| Parameter-based proposals | Providers “may propose to individual investors specific crowdfunding projects that correspond to one or more specific parameters or risk indicators chosen by the investor” [S17] | A close analogue to mandate-based filtering |
| Conflicts of interest | Includes a prohibition on investing in offers on the provider’s own platform, and effective internal conflict rules [S17][S18] | Supports the separation of platform and agent capital |
| Individual portfolio management | Discretion within investor-set parameters is permitted for *loans*, with disclosure of the decision-making process [S17] | Discretionary agent allocation of *equity* is not covered by this derogation |
| Investor protection | Entry knowledge test and loss-bearing simulation for non-sophisticated investors. A Key Investment Information Sheet (KIIS) for each offer [S18][S19] | Onboarding gates in the policy engine |
| Due diligence | Minimum due diligence on project owners [S19] | Provided by the Evidence Engine |
| Record keeping | Records kept for at least five years [S18] | Satisfied by the decision ledger |

## 5.3 Analysis

The ECSPR’s investor-chosen parameter mechanism is structurally similar to Valthera’s mandates. Its discretionary-management derogation, however, covers loan portfolios only. Agent-driven equity allocation for third-party investors in the EU would therefore likely need authorization beyond crowdfunding (for example, under MiFID II portfolio management). The alternative is to restrict agents to deploying their operators’ own capital. *Confidence: Medium.* The regulation text is primary; its application to AI agents has not been tested.

## 5.4 Jurisdictions not analyzed

The UK (FCA), Singapore (MAS), the UAE (VARA, ADGM, DIFC), and Switzerland were not analyzed. The EU Markets in Crypto-Assets Regulation (MiCA) was not analyzed in detail.

# 6. RQ5 — Security Risks of Agents That Control Capital

## 6.1 Authoritative risk framework

The OWASP Top 10 for LLM Applications (2025) ranks prompt injection first (LLM01) [S21]. OWASP states that injected inputs can affect a model "even if they are imperceptible to humans." Among its listed mitigations [S21]:

- restrict model privileges "to the minimum necessary for its intended operations";
- "implement human-in-the-loop controls for privileged operations";
- "separate and clearly denote untrusted content";
- handle extensible functions "in code rather than providing them to the model";
- perform penetration testing that treats "the model as an untrusted user."

One OWASP scenario is directly relevant. A résumé containing split malicious prompts manipulates an LLM evaluator into a positive recommendation [S21]. Pitch decks submitted to an AI diligence system present the same attack surface. The 2025 list also includes "excessive agency": systems that grant models more functionality, permissions, or autonomy than they need [S21-list].

## 6.2 Documented incident: Grok / Bankrbot (May 2026)

The OECD AI Incidents Monitor records that an attacker exploited the AI agents Grok and Bankrbot "by sending a Morse code prompt via X, tricking them into transferring 3 billion DRB tokens" worth $150,000–$200,000 from a wallet on Base [S22]. MetaMask's security report describes it as the first documented prompt-injection attack against an AI agent with real financial capabilities. It notes that no smart-contract vulnerability or key compromise was involved, and that an NFT sent to the wallet had expanded its permissions [S23].

**Attack chain**

1. An inbound asset (a membership NFT) expanded the wallet's execution permissions.
2. An encoded instruction was posted publicly.
3. One AI system decoded it, and a second treated the decoded text as an authorized command.
4. A standard token transfer executed with no human approval.

**Discrepancies.** Reported loss figures range from about $150,000 to $204,000 across sources. Reports that 80–88% of funds were returned are not confirmed by official statements [S30].

## 6.3 Control mapping

| Attack step | Valthera control | Whitepaper reference |
|---|---|---|
| Inbound asset expands permissions | Permissions change only through signed mandate updates with a timelock. Inbound assets have no effect on authority | §10.3, §11.3 |
| Untrusted text reaches the model | Channel separation, wrapping of untrusted content, injection detection including encoded payloads | §16.3 |
| Model output treated as a command | Analysis models have no tools that change state. Outputs must match a schema and are never executed | §16.3 |
| Transfer executes without review | Policy engine, onchain vault limits, allowlisted counterparties, human approval above thresholds | §11.2–11.3 |

*Confidence: High (framework); Medium (incident specifics).*

# 7. RQ6 — Measuring Agent Skill and Performance

## 7.1 The distribution of early-stage returns

AngelList analyzed 1,808 pre–Series C investments made through its platform. It found that early-stage returns “follow a fairly extreme power law distribution, with enormous positive outliers skewing portfolio performance,” with a fitted tail exponent (α) of about 2.3 [S27]. It also found that indexing the full early-stage universe would outperform roughly three-quarters of early-stage venture funds [S27].

**Implications for measurement**

- A small portfolio’s returns are dominated by whether it held an outlier, so short-run returns are a noisy measure of skill.
- Outcomes take years to resolve, which delays return-based evaluation.
- The heavy tail means that ranking agents on raw returns would reward luck as much as skill.

A frequently cited Correlation Ventures statistic (about 65% of deals returning under 1x, about 4% returning over 10x) was found only in secondary sources. It is excluded from this report and from the whitepaper.

## 7.2 Proper scoring rules for forecasts

The Brier score was introduced by Glenn W. Brier in 1950 to verify probabilistic weather forecasts (*Monthly Weather Review* 78(1):1–3). It is the mean squared difference between forecast probabilities and binary outcomes. Lower is better, and 0 is perfect [S25]. It is a *strictly proper* scoring rule: a forecaster minimizes their expected score by reporting their true belief [S25][S26]. Strictly proper scoring rules are formally characterized in Gneiting and Raftery (2007) [S26].

**Known limitations**

- The Brier score depends on prevalence, so the ranking of forecasters can shift with how common the event is [S31].
- For rare events, it discriminates poorly between forecasts that differ in ways that matter, so large samples are needed to tell skill levels apart.

**Implications for design**

- Score agents only on shared question sets.
- Use skill scores relative to base rates.
- Require minimum sample sizes before publishing rare-event scores.
- Supplement Brier scores with log scores and calibration diagnostics.

## 7.3 Resulting measurement framework

| Dimension | Metric | Resolution speed | Noise |
|---|---|---|---|
| Forecast skill | Brier skill score, log score, calibration | Months (milestones, follow-ons) | Low–moderate |
| Selection | Skill on invest/decline forecasts | Months to years | Moderate |
| Pricing | Entry valuation compared with clearing valuation and later marks | Years | High |
| Returns | DPI, TVPI, IRR, PME | Years | Very high (power law) |

*Confidence: High.*

# 8. Design Implications Matrix

| # | Finding | Grade | Design implication | Whitepaper v1.0 |
|---|---|---|---|---|
| 1 | Agent autonomy rising, mostly on software tasks | High / Low transfer | Do not assume diligence competence. Start with human review | §1, §22 |
| 2 | Agent payment volume small, partly artificial | Low–Medium | Treat rails as interchangeable. Avoid demand claims | §2.4, §13.3 |
| 3 | LLM financial accuracy limited, errors look confident | Medium | Models propose, engines compute. Missing data stays missing. Audit sampling | §6, §8.2 |
| 4 | Funding portals cannot recommend or hold funds | High | Separate research from offerings. Broker-dealer route for US | §20.2 |
| 5 | Intermediary gatekeeper duties (Rule 301) | High | Evidence Engine signals feed partner due diligence | §6, §20 |
| 6 | Publisher exclusion requires impersonal content | Medium | Uniform assessments. No portfolio-tailored advice from the platform | §15, §20 |
| 7 | Auto-trading falls outside the publisher exclusion | Medium | Operators of executing agents carry their own registration | §11, §20.4 |
| 8 | SEC AI conflict proposal withdrawn | High | Keep voluntary conflict registry and disclosures | §15.3 |
| 9 | Howey-based token taxonomy; tokenized securities remain securities | Medium | No token. Instruments treated as securities | §12.6, §21 |
| 10 | ECSPR: €5M scope, EU establishment, no inducements | Medium–High | EU subsidiary for EU capital formation. Usage-based fees only | §20.3, §21 |
| 11 | ECSPR discretion limited to loans | Medium | EU equity discretion requires further authorization | §20.3 |
| 12 | Prompt injection ranked first; real fund-draining incident | High / Medium | Capability separation, controls outside the model, human approval | §11, §16 |
| 13 | Power-law returns | Medium–High | Separate forecast skill from returns. Risk and vintage adjustment | §14.5 |
| 14 | Proper scoring rules depend on prevalence | High | Shared question sets, skill scores, minimum sample sizes | §14.3 |

# 9. Confidence and Gaps

## 9.1 Confidence by area

| Area | Confidence | Basis |
|---|---|---|
| US offering exemptions and limits | High | SEC primary sources |
| Funding-portal prohibitions | High | SEC and FINRA primary sources, consistent |
| Investment Advisers Act publisher exclusion | Medium | Supreme Court primary; Seeking Alpha via law-firm summaries; untested for AI agents |
| Withdrawal of the SEC AI conflict proposal | High | Federal Register |
| SEC token taxonomy | Medium | Chairman's speeches with disclaimers; final text not reviewed |
| EU ECSPR provisions | Medium–High | EUR-Lex primary; application to agents untested |
| METR capability trend | Medium–High (trend), Low (transfer) | Primary paper with stated limits |
| LLM financial accuracy | Medium | Primary benchmark papers; model generations change quickly |
| Agent payment volumes | Low–Medium | News report of third-party analysis; conflicting figures |
| Prompt-injection framework | High | OWASP primary |
| Grok/Bankrbot incident | Medium | OECD and MetaMask confirm mechanism; loss figures vary; fund return unverified |
| Venture return distribution | Medium–High | AngelList primary dataset (platform-specific sample) |
| Scoring-rule properties | High | Established statistical literature |

## 9.2 Source conflicts and how they were resolved

| Topic | Conflict | Resolution |
|---|---|---|
| x402 activity | Hundreds of millions of transactions vs. about $28,000 daily dollar volume | Dollar volume preferred as the measure of economic activity. Counts include sub-cent and artificial transactions |
| Grok/Bankrbot loss | $150K, $155–180K, $175K, $204K reported | A range is reported. Differences likely reflect token price at different times |
| Grok/Bankrbot recovery | 80–88% reportedly returned | Treated as unverified. The source itself notes no official confirmation |
| METR doubling time | 7 months (2019–2025) vs. faster recent estimates | The longer-window estimate is used. Shorter windows are less robust, as METR notes |

## 9.3 Gaps

1. **AI efficacy in investment selection.** No rigorous study was located showing AI agents outperform people at early-stage selection or valuation as measured by outcomes. This is the thesis's central unproven assumption. Valthera's scoring framework is designed to test it.
2. **Other jurisdictions.** UK, Singapore, UAE, and Swiss regimes, and MiCA in detail, were not analyzed.
3. **State-level US requirements.** State securities ("blue sky") notice filings and state adviser registration were not analyzed.
4. **Primary texts not reviewed.** The Seeking Alpha opinion, the March 2025 SEC no-action letter, the DERA report, and the final SEC token interpretation were relied on through secondary summaries.
5. **Competitive landscape.** Existing AI diligence products, agent-investing protocols, and crowdfunding platforms were not systematically surveyed.
6. **Founder demand.** No evidence was gathered on founders' willingness to submit to public, evidence-labeled

assessment.

7. **Market sizing.** No addressable-market estimate is made, and none is claimed in the whitepaper.
8. **Tax treatment.** The tax implications of agent-held positions and cross-border instruments were not analyzed.

# 10. Source Register

| ID | Source | Type |
|---|---|---|
| S1 | METR, “Measuring AI Ability to Complete Long Tasks” (2025). https://metr.org/blog/2025-03-19-measuring-ai-ability-to-complete-long-tasks/ ; paper: https://arxiv.org/abs/2503.14499 | Primary research |
| S2 | Coinbase, “Introducing x402.” https://www.coinbase.com/developer-platform/discover/launches/x402 | Primary (developer) |
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